Complex Analysis

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Chapter 1

Complex Numbers

A complex number is an expression of the form $z = x + iy$, where x and y are real numbers. The component x is called the real part of z, and y is the imaginary part of z. We will denote these by

$$x = \operatorname{Re} z,$$
$$y = \operatorname{Im} z.$$

The set of complex numbers forms the complex plane, which we denote by $\mathbb{C}$. We denote the set of real numbers by $\mathbb{R}$, and we think of the real numbers as being a subset of the complex plane, consisting of the complex numbers with imaginary part equal to zero.

The correspondence

$$z = x + iy \longleftrightarrow (x, y)$$

is a one-to-one correspondence between complex numbers and points (or vectors) in the Euclidean plane $\mathbb{R}^2$. The real numbers correspond to the x-axis in the Euclidean plane. The complex numbers of the form iy are called purely imaginary numbers. They form the imaginary axis $i\mathbb{R}$ in the complex plane, which corresponds to the y-axis in the Euclidean plane.

1.0.1 Basic Definitions and Algebra

- A complex number has the form $z = a + ib$ where a, b are real numbers
- a is the **real part**: $\operatorname{Re}(z) = a$
- b is the **imaginary part**: $\operatorname{Im}(z) = b$
- A number is **purely imaginary** if $a = 0$
- A number is **real** if $b = 0$
- Zero is the only number that is both real and purely imaginary

1.0.2 Arithmetic Operations

1.0.2.1 Addition

$(a+ib)+(c+id)=(a+c)+i(b+d)$

1.0.2.2 Multiplication

$(a+ib)(c+id)=(ac-bd)+i(ad+bc)$

1.0.2.3 Division

$$\frac{a+ib}{c+id}=\frac{(a+ib)(c-id)}{(c+id)(c-id)}=\frac{ac+bd}{c^2+d^2}+i\frac{bc-ad}{c^2+d^2}$$

1.0.3 Complex Conjugate

- The conjugate of $z=a+ib$ is $\bar{z}=a-ib$
- Properties:
  - $\overline{z_1+z_2}=\bar{z}_1+\bar{z}_2$
  - $\overline{z_1\cdot z_2}=\bar{z}_1\cdot\bar{z}_2$
  - $\overline{\left(\frac{z_1}{z_2}\right)}=\frac{\bar{z}_1}{\bar{z}_2}$
  - z is real if and only if $z=\bar{z}$
  - z is purely imaginary if and only if $z=-\bar{z}$
- Expressing real and imaginary parts using conjugate:
  - $\mathrm{Re}(z)=\frac{z+\bar{z}}{2}$
  - $\mathrm{Im}(z)=\frac{z-\bar{z}}{2i}$

1.0.4 Absolute Value (Modulus)

- The absolute value or modulus of $z=a+ib$ is:
  - $|z|=\sqrt{a^2+b^2}=\sqrt{z\bar{z}}$
- Properties:
  - $|z|\geq 0$ for all z, with equality if and only if $z=0$
  - $|z_1\cdot z_2|=|z_1|\cdot|z_2|$
  - $\left|\frac{z_1}{z_2}\right|=\frac{|z_1|}{|z_2|}$ for $z_2\neq 0$
  - $|z_1+z_2|\leq|z_1|+|z_2|$ (Triangle inequality)
  - $||z_1|-|z_2||\leq|z_1-z_2|$

1.0.5 Important Inequalities

1. **Triangle Inequality**: $|z_1 + z_2| \leq |z_1| + |z_2|$
   - Equality holds if and only if $\frac{z_1}{z_2} > 0$ (i.e., the ratio is positive real)
2. $-|z| \leq \text{Re}(z) \leq |z|$
   - Equality on right when z is positive real
3. $-|z| \leq \text{Im}(z) \leq |z|$
4. **Cauchy's Inequality**: $|a_1b_1 + a_2b_2 + ... + a_nb_n|^2 \leq (|a_1|^2 + ... + |a_n|^2)(|b_1|^2 + ... + |b_n|^2)$
   - Equality holds if and only if the a_i are proportional to the b_i

1.1 Geometric Representation

1.1.1 The Complex Plane

- A complex number $z = a + ib$ corresponds to the point (a, b) in the plane
- The x-axis is called the **real axis**
- The y-axis is called the **imaginary axis**
- The plane is called the **complex plane**

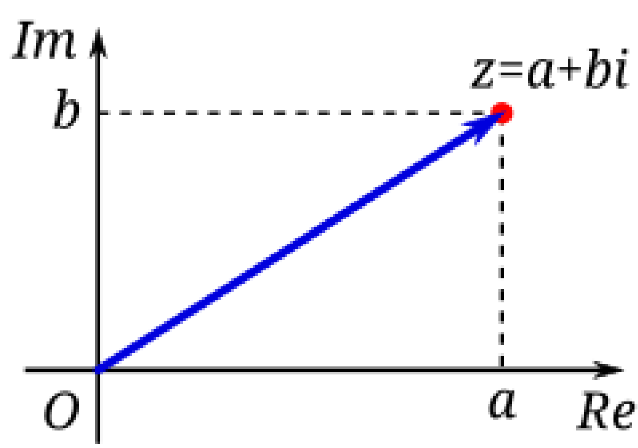


Figure 1.1: Image Source: Wikimedia

1.1.2 Vector Interpretation

- Complex numbers can be represented as vectors from the origin
- Vector addition corresponds to complex addition
- The length of the vector is $|z|$
- The distance between points z_1 and z_2 is $|z_1 - z_2|$

1.1.3 Polar Form

- A complex number can be written in **polar form**:
  - $z = r(\cos\theta + i\sin\theta)$ where $r = |z|$ and $\theta = \arg(z)$
  - r is the modulus: $r = |z| = \sqrt{a^2 + b^2}$
  - θ is the **argument** or **amplitude**: $\theta = \arg(z)$
- Properties of polar form:
  - For $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$:
  - $z_1 z_2 = r_1 r_2[\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2)]$
  - $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$
  - $\arg\left(\frac{z_1}{z_2}\right) = \arg(z_1) - \arg(z_2)$

1.1.4 De Moivre's Formula

- For $z = r(\cos\theta + i\sin\theta)$:
  - $z^n = r^n(\cos n\theta + i\sin n\theta)$
- For $r = 1$:
  - $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$

1.1.5 Roots of Complex Numbers

- The n-th roots of $z = r(\cos\theta + i\sin\theta)$ are:
  - $\sqrt[n]{z} = \sqrt[n]{r}\left[\cos\left(\frac{\theta+2\pi k}{n}\right) + i\sin\left(\frac{\theta+2\pi k}{n}\right)\right]$ for $k = 0, 1, 2, \ldots, n-1$
- There are n distinct n-th roots of any non-zero complex number
- These roots have equal moduli and are equally spaced around a circle
- For $z = 1$, the roots are called the **n-th roots of unity**

1.1.6 Equations in the Complex Plane

- Circle with center a and radius r: $|z-a|=r$
  - In algebraic form: $(z-a)(\bar{z}-\bar{a})=r^2$
- Straight line: $z=a+bt$ where t is a real parameter
  - Two lines are parallel if b' is a real multiple of b
  - Lines are orthogonal if $\frac{b'}{b}$ is purely imaginary
  - The angle between lines is $\arg\frac{b'}{b}$

1.1.7 Stereographic Projection and the Riemann Sphere

The extended complex plane is the complex plane together with the point at infinity. We denote the it by $C^* = c \cup \{\infty\}$. - The extended complex plane includes ∞ and can be represented on a unit sphere - Under stereographic projection, the point (x_1, x_2, x_3) on the sphere corresponds to: - $z=\frac{x_1+ix_2}{1-x_3}$

- The north pole $(0,0,1)$ corresponds to ∞
- Properties:
  - Circles on the sphere correspond to circles or lines in the plane
  - The distance formula: $d(z,z')=\frac{2|z-z'|}{\sqrt{(1+|z|^2)(1+|z'|^2)}}$
  - The sphere is often called the **Riemann sphere**

1.1.8 Key Formulas and Results

1. $z\bar{z}=|z|^2=a^2+b^2$
2. $\frac{1}{z}=\frac{\bar{z}}{|z|^2}=\frac{a-ib}{a^2+b^2}$
3. $z^n=r^n(\cos n\theta+i\sin n\theta)$
4. $\sqrt[n]{z}=\sqrt[n]{r}\left[\cos\left(\frac{\theta+2\pi k}{n}\right)+i\sin\left(\frac{\theta+2\pi k}{n}\right)\right]$ where $k=0,1,\ldots,n-1$
5. Triangle inequality: $|z_1+z_2|\le|z_1|+|z_2|$
6. Cauchy's inequality: $\left|\sum_{i=1}^n a_i\bar{b}_i\right|^2\le\left(\sum_{i=1}^n|a_i|^2\right)\left(\sum_{i=1}^n|b_i|^2\right)$
7. The equation of a circle: $|z-a|=r$
8. The roots of unity: ω^k where $\omega=\cos\frac{2\pi}{n}+i\sin\frac{2\pi}{n}$ and $k=0,1,\ldots,n-1$

Chapter 2

Complex Functions

Chapter 3

Exercise

1. Find the values of

$$(1+2i)^3, \quad \frac{5}{-3+4i}, \quad \left(\frac{2+i}{3-2i}\right)^2, \quad (1+i)^n + (1-i)^n.$$

2. If $z = x+iy$ (where x and y are real numbers), find the real and imaginary parts of:

$$z^4, \quad \frac{1}{z}, \quad \frac{z-1}{z+1}, \quad \frac{1}{z^2}.$$

3. Show that:

$$\left(\frac{-1 \pm i\sqrt{3}}{2}\right)^3 = 1 \quad \text{and} \quad \left(\frac{\pm 1 \pm i\sqrt{3}}{2}\right)^6 = 1$$

for all combinations of signs.

Chapter 4

Green functions

Let G be a reigon and $a \in G$. Green functions G with singularity at a, the unique function $g_a : G \setminus \{0\}$

1. g_a is harmonisc on $G {a}.
2. $g_a(z) + \log(|z - a|)$ is harmonic[1] on an open neighbourhood of a.
3. $\lim_{z \to w} g_a(z) = 0 \quad \forall w \in \partial_\infty G$.

::: {.remark} $g_a(z) > 0$ for all $z \in G \setminus \{a\}$ because $g_a(z) + \log(|z - a|)$ has contious externsion to u and lim_{z \to w} g_a(z) = -\infty$ :::

We have $g_a(z)0$ on $|z - a| = r$ for r small enough (then there exist R such that its positive $\forall r < R$) Given $r < R$ such that z_0 is $|z - a| = r$. Apply max principle on $G \setminus \{z : |z - a| \leq r\}$. Since $g_a(z) \geq 0$ or $$

[1]has an exteror to u. $(a \in u)$, u is open in G.

Chapter 5

Riemann Surfaces.

Definition 5.1 (Riemann surface)**.** A **Riemann surface** is a second-countable, Hausdorff topological space X equipped with a **complex atlas** $\mathcal{A} = \{(U_\alpha, \varphi_\alpha)\}$ satisfying the following properties:

1. Each $U_\alpha \subseteq X$ is open.
2. Each map $\varphi_\alpha : U_\alpha \to V_\alpha$ is a homeomorphism onto an open set $V_\alpha \subseteq \mathbb{C}$.
3. The collection $\{U_\alpha\}$ covers X.
4. For every pair of charts $(U_\alpha, \varphi_\alpha)$ and (U_β, φ_β) with $U_\alpha \cap U_\beta \neq \emptyset$, the **transition map**

$$\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \longrightarrow \varphi_\beta(U_\alpha \cap U_\beta)$$

   is a **biholomorphism**.

Remark. **Terminology**: -"holomorphic functions" are functions which are complex differentiable on an open set in the complex plane. -A "biholomorphic map" is a holomorphic bijection whose inverse is holomorphic.

Definition 5.2. Let R_1 and R_2 be Riemann surfaces. A map $f : R_1 \to R_2$ is said to be **holomorphic** if for every pair of charts $(U, \varphi) \quad \text{on } R_1, \qquad (V, \psi) \quad \text{on } R_2,$ the map $\psi \circ f \circ \varphi^{-1}$ is holomorphic on its domain (which may be empty).

Suppose (U, φ), (U', φ') are charts on R_1, and (V, ψ), (V', ψ') are charts on R_2. On the overlap where all maps are defined, we have:

- the transition maps $\varphi' \circ \varphi^{-1}, \psi' \circ \psi^{-1}$ are **biholomorphisms** (by condition (5) in the definition of a complex atlas)

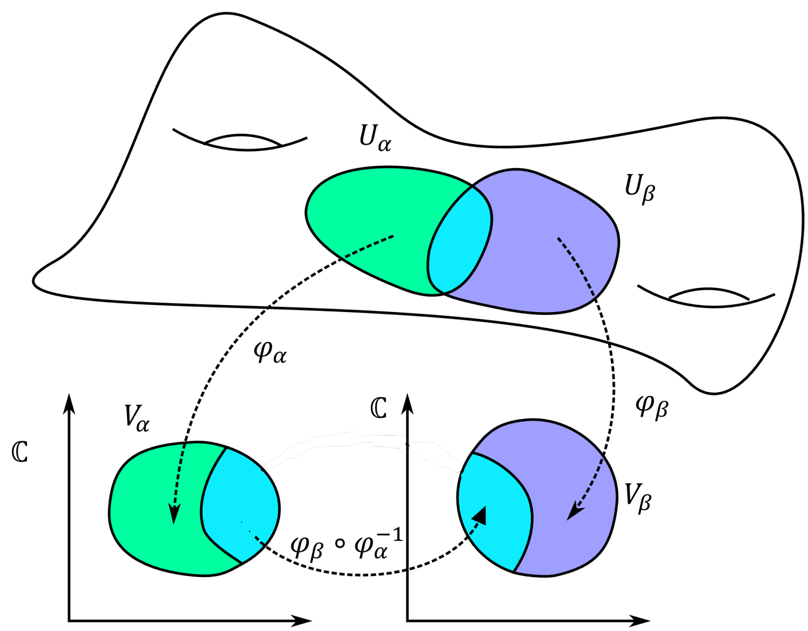


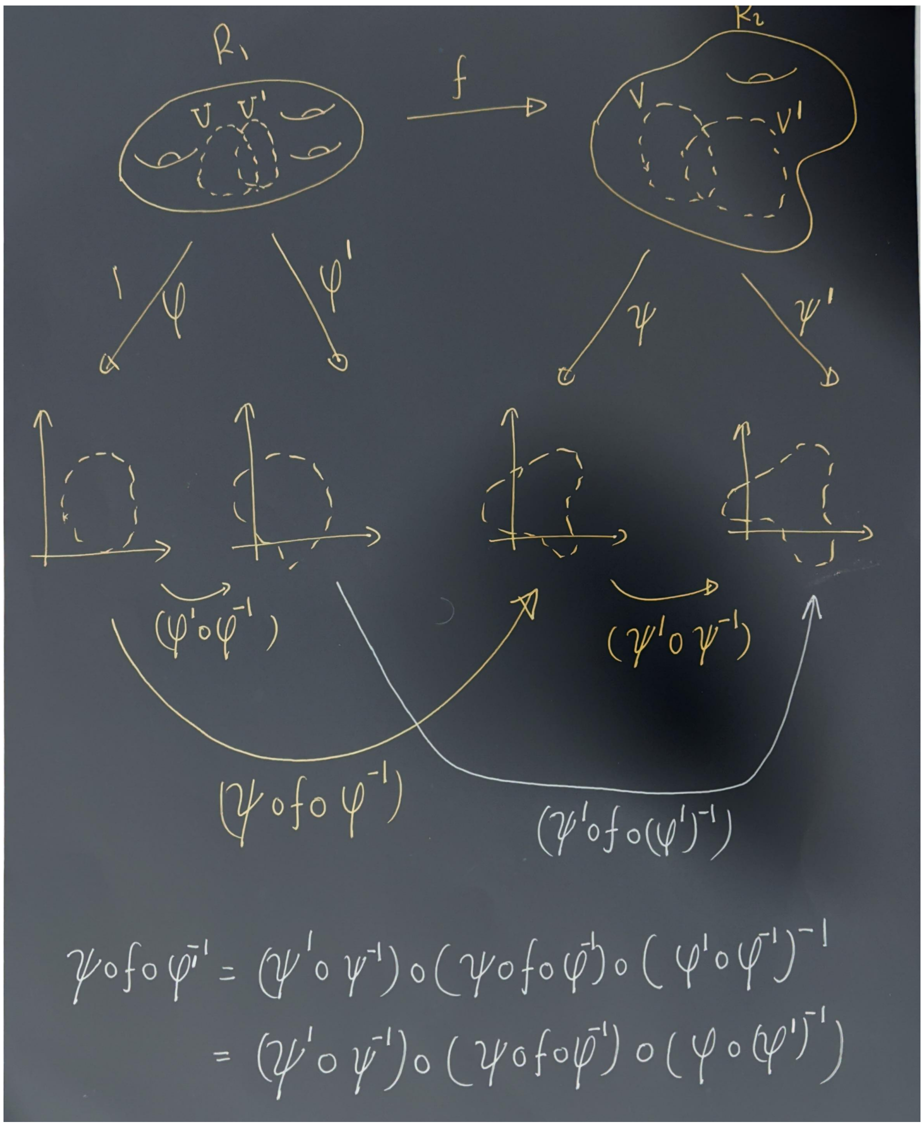
Figure 5.1: Two coordinate charts on a manifold (Image Credit: Wikipedia)

- therefore

$$\psi' \circ f \circ (\varphi')^{-1} = (\psi' \circ \psi^{-1}) \circ (\psi \circ f \circ \varphi^{-1}) \circ (\varphi \circ (\varphi')^{-1})$$

  is holomorphic if and only if $\psi \circ f \circ \varphi^{-1}$ is holomorphic.

Thus the notion of holomorphicity does **not** depend on the choice of charts. "'



Exercise 5.1. Assume that for all $p \in R_1$ there is a chart (U, ϕ) such that $p \in U$ and a chart (V, ψ) such that $f(p) \in V$, so that $\psi \circ f \circ \phi^{-1}$ is holomorphic

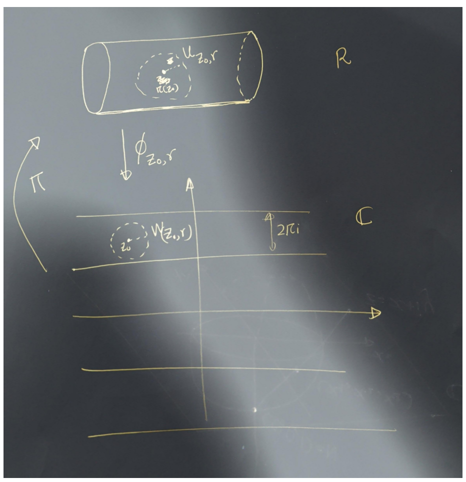
(in the standard sense, as a complex analytic function on an open subset of the plane). Show that f is holomorphic.

Solution:I will update it later.

Example 5.1. Define the equivalence relation on $\mathbb{C}$ by $z \sim w \quad$ if and only if $\quad z = w + 2\pi i n \quad$ for some $n \in \mathbb{Z}$. Let $R = \mathbb{C}/\sim$ as a topological space, with projection map $\pi : \mathbb{C} \to R$.

We will define an atlas on R which will make it a Riemann surface. Define $W_{z_0,r} = \{ z \in \mathbb{C} : |z - z_0| < r \}$. Restrict to $r < \pi$ (for convenience). Let $U_{z_0,r} = \pi(W_{z_0,r})$.

Let $\varphi_{z_0,r} : U_{z_0,r} \to W_{z_0,r}$ be the unique right inverse of π on $U_{z_0,r}$ with image $W_{z_0,r}$.



Exercise 5.2. Show that for any $p \in W$ and any $q \in W$, the map $\sigma(q) - \sigma(p) : U \longrightarrow \mathbb{C}$ is a **translation**, and in particular holomorphic.

5.1 The Riemann sphere

As a set, the **Riemann sphere** is $\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$. To make this into a Riemann surface, we must specify a **topology** and an **atlas**.

A basis for the topology consists of:

1. Sets of the form $U \subset \mathbb{C}$, where U is open in $\mathbb{C}$ and $\infty \notin U$.
2. Sets of the form $\{\infty\} \cup (\mathbb{C} \setminus K)$, where $K \subset \mathbb{C}$ is compact.

Equivalently, a set $V \subset \widehat{\mathbb{C}}$ is open iff:

- If $p \in \mathbb{C}$, then there exists an open set $U \subset \mathbb{C}$ with $p \in U \subset V$.
- If $p = \infty$, then there exists a compact $K \subset \mathbb{C}$ such that $\{\infty\} \cup (\mathbb{C} \setminus K) \subset V$.

Exercise 5.3. Show that $\widehat{\mathbb{C}}$ is compact.

We define an atlas consisting of two charts:

$$U_1 = \widehat{\mathbb{C}} \setminus \{\infty\}, \qquad \varphi_1(z) = z.$$
$$U_2 = \widehat{\mathbb{C}} \setminus \{0\}, \qquad \varphi_2(z) = \frac{1}{z}.$$

The first chart and its inverse are clearly continuous.

To see that φ_2 is continuous, note that it maps $U_1 \cap U_2 = \widehat{\mathbb{C}} \setminus \{0, \infty\}$ bijectively onto $\mathbb{C} \setminus \{0\}$.

The overlap maps are:

$$\varphi_2 \circ \varphi_1^{-1}(z) = \frac{1}{z}, \qquad \varphi_1 \circ \varphi_2^{-1}(w) = \frac{1}{w}.$$

These are biholomorphic, since the inverse of $z \mapsto 1/z$ is again $z \mapsto 1/z$.

Thus, this defines a valid atlas, and the resulting Riemann surface is the **Riemann sphere**.

5.1.1 Stereographic Projection and the Metric

We identify the Riemann sphere with the unit sphere $S^2 = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1^2 + x_2^2 + x_3^2 = 1\}$. Think of $\mathbb{C}$ as the plane $(x_1, y_1, 0)$. Given $z = x + iy \in \mathbb{C}$, draw the line from the north pole $(0, 0, 1)$ to $(x, y, 0)$.
This intersects the sphere at:

$$x_1 = \frac{2x}{|z|^2 + 1} = \frac{2x}{x^2 + y^2 + 1},$$
$$x_2 = \frac{2y}{|z|^2 + 1} = \frac{2y}{x^2 + y^2 + 1},$$
$$x_3 = \frac{|z|^2 - 1}{|z|^2 + 1} = \frac{x^2 + y^2 - 1}{x^2 + y^2 + 1}.$$

Define

$$\sigma : S^2 \to \widehat{\mathbb{C}}$$

by

$$\sigma(x_1, x_2, x_3) = \begin{cases} \dfrac{x_1}{1-x_3} + i\,\dfrac{x_2}{1-x_3}, & (x_1, x_2, x_3) \neq (0,0,1), \\ \infty, & (x_1, x_2, x_3) = (0,0,1). \end{cases}$$

$$\sigma^{-1}(z) = \begin{cases} \left(\dfrac{2\Re z}{|z|^2+1}, \dfrac{2\Im z}{|z|^2+1}, \dfrac{|z|^2-1}{|z|^2+1} \right), & z \in \mathbb{C}, \\ (0,0,1), & z = \infty. \end{cases}$$

The spherical (or chordal) metric is:

$$d_{\widehat{\mathbb{C}}}(z,w)| = \frac{2|z-w|}{\sqrt{(1+|z|^2)(1+|w|^2)}},$$

with the natural extension when one point is ∞.